

FACULTY OF INFORMATICS

**B.C.A. V Semester (CBCS) (Main & Backlog & One Time Chance) Examination,
February 2023**

Subject: Data Mining

Time: 3 Hours

Max. Marks: 70

**Note: I. Note: Answer all questions from Part-A and answer any five questions from Part-B. Choosing one question from each unit.
II. Missing data, if any, may be suitably assumed.**

PART - A (10 x 2 = 20 Marks)

1. a) Define Data mining.
b) Discuss about Outlier Analysis.
c) What is Market Basket analysis? Discuss its importance.
d) What are multilevel association rules?
e) What is Gini Index?
f) Explain the working principle of SVM.
g) What are different types of data in Cluster analysis?
h) Define Constraint-based Clustering.
i) What is Discriminant analysis?
j) Define Web-wide tracking.

PART -B (5 x 10 = 50 Marks)

Unit-I

2. a) Discuss the steps in the Knowledge discovery process.
b) Discuss about Data mining task primitives.
(OR)
3. a) Discuss the major issues in Data Mining.
b) Explain the major steps involved in data preprocessing.

Unit-II

4. a) Describe frequent pattern mining.
b) Explain Apriori algorithm.
(OR)
5. a) Define horizontal data format and Vertical data formats.
b) Explain the process of mining frequent itemsets using the vertical data format with an example.

Unit-III

6. a) Compare Classification and Prediction methods.
b) What is Tree Pruning? Explain how does tree pruning work.
(OR)
7. a) What are Bayesian classifiers? Discuss
b) Explain Bayesian Belief Networks with an example.

Unit-IV

8. a) Discuss about Categorical, Ordinal, and Ratio-Scaled Variables.
b) Explain about K-means method.

(OR)

9. a) Define Hierarchical clustering.
b) Discuss about Hierarchical clustering methods.

Unit-V

10. a) Discuss the applications of Data mining for Biological Data Analysis.
b) Discuss about Visual and Audio Data Mining

(OR)

11. a) Explain about Statistical Data Mining.
b) Discuss about privacy preserving data mining methods.

Code No: E-15334/CBCS

FACULTY OF INFORMATICS
B.C.A. V - Semester (CBCS) (Main & Backlog & One Time Chance) Examination,
February 2023

Time: 3 Hours

Subject: Software Quality and Testing

Max. Marks: 70

Note: I. Note: Answer all questions from Part-A and answer any five questions from Part-B. Choosing one question from each unit.
II. Missing data, if any, may be suitably assumed.

PART - A (10 x 2 = 20 Marks)

1. a) Define Quality
- b) What is Quality control?
- c) What are the characteristics of Inspection?
- d) Describe about Feature Point.
- e) What is Defect?
- f) Describe Unit testing.
- g) What is Debugging?
- h) Describe about Test Log.
- i) What is a Walkthrough?
- j) What is a Review?

PART-B (5 x 10 = 50 Marks)

Unit-I

2. a) Discuss about Quality Standards in detail.
 - b) Explain the Characteristics of Information Engineering.
- (OR)
3. a) Discuss the Techniques of Measuring Customer Satisfaction.
 - b) Explain about Software Quality Engineering in detail.

Unit-II

4. a) Explain about Software Reliability Metrics in detail.
 - b) Discuss about Agile Process Metrics in detail.
- (OR)
5. a) Explain about Lines of Code (LOC) in detail.
 - b) Explain various Software Cost Estimation models.

Unit-III

6. a) Discuss the Role of Software Testing in SDLC.
 - b) Explain the Techniques for White-Box Testing.
- (OR)
7. a) Discuss about Evaluating Test Adequacy Criteria.
 - b) Explain about Integration Testing in detail.

Unit-IV

8. a) Explain the Principles of Software Testing in brief.
b) Discuss about Test Case Specifications in detail.

(OR)

9. a) Discuss about Test Summary Report in detail.
b) Explain about Process and the Engineering.

Unit-V

10. a) Explain the Functions of Project Controlling and Monitoring.
b) Discuss the Criteria for Test Completion in detail.

(OR)

11. a) Explain about Software Configuration Management.
b) Discuss the Phases of Developing Formal Review.

Subject: Object Oriented System Development

Time: 3 Hours

Max. Marks: 70

Note: I. Note: Answer all questions from Part-A and answer any five questions from Part-B. Choosing one question from each unit.

II. Missing data, if any, may be suitably assumed.

PART - A (10 x 2 = 20 Marks)

1. a) Define class.
b) List the relationships used in class diagram.
c) What is swimlane in Activity Diagram?
d) Define substate.
e) What is Artifact?
f) Define collaboration.
g) List four p's of unified process.
h) What are the benefits of Iterative Process?
i) Define Analysis phase.
j) List the workers involved in the Requirement workflow.

PART - B (5 x 10 = 50 Marks)

Unit-I

2. a) What is UML? What is the importance of modeling?
b) What are the building blocks of UML?

(OR)

3. a) Explain about Object diagram.
b) Give an example of Object diagram.

Unit-II

4. a) What is Usecase diagram? Explain the notations used in Usecase diagram.
b) Give an example of Usecase diagram.

(OR)

5. a) Write about Process & Threads in UML.
b) Explain about Events & Signals in State Machine.

Unit-III

6. a) Explain about Deployment diagram.
b) Give any example for Deployment diagram.
7. a) Explain about Patterns & Frameworks.
b) Give an example of collaboration diagram.

Unit-IV

8. a) Explain about Unified process.
b) Write short note on Use case driven process.
(OR)
9. a) Explain about Iterative & Incremental process.
b) Briefly explain about Architecture centric process.

Unit-V

10. a) What are the activities in the Requirement core workflow of unified process.
b) Explain the Requirement gathering using Usecase model.
(OR)
11. a) Explain the activities in the Design core workflow.
b) Explain the activities in the Test core workflow.

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FACULTY OF INFORMATICS
B.C.A. V - Semester (CBCS) (Main & Backlog & One Time Chance) Examination,
February 2023

Time: 3 Hours

Subject: UNIX Programming

Max. Marks: 70

Note: I. Note: Answer all questions from Part-A and answer any five questions from Part-B. Choosing one question from each unit.

II. Missing data, if any, may be suitably assumed.

PART – A (10 x 2 = 20 Marks)

1. a) Discuss different types of files in Unix.
- ~~b) What is a Regular expression?~~
- ~~c) What are pipes? What are its limitations?~~
- ~~d) What is the difference between wait and Waitpid function?~~
- ~~e) Difference between poll function and select function.~~
- ~~f) What is the purpose of bind and connect?~~
- ~~g) Write about compound data types in PHP.~~
- ~~h) What is the use GET and POST methods in form handling?~~
- ~~i) Differentiate between break and continue statements.~~
- ~~j) What are anonymous functions in Python?~~

PART – B (5 x 10 = 50 Marks)

Unit-I

2. a) Explain the salient features of Unix Operating System.
 - ~~b) What are file permissions? Explain how the file permissions can be given or revoked.~~
- (OR)
3. a) Write short notes on Shell programming.
 - b) Discuss about awk with suitable examples.

Unit-II

4. a) Define Signal. Explain different methods to handle signals.
 - ~~b) Discuss about locking techniques in Unix.~~
- (OR)
5. a) Define FIFO. Discuss how it is useful in IPC.
 - ~~b) Explain about Shared memory and its system calls.~~

Unit-III

6. a) Describe advanced socket system calls.
 - b) Elaborate the concept of reserved ports.
- (OR)
7. a) Explain about Input/Out multiplexing.
 - ~~b) Discuss about Internet super server.~~

Unit-IV

8. a) Explain control statements in PHP with suitable examples.
b) What are the different ways to create an array in PHP? Explain with an example.
- (OR)
9. a) Discuss briefly about cookies and session tracking in PHP.
b) Explain about file handling in PHP with an example.

Unit-V

10. a) Define list. Explain various list methods with suitable examples.
b) Explain, with examples, the various Python looping statements.
- (OR)
11. a) Demonstrate the use of Exception Handling in Python.
b) What is Inheritance? Explain the concept of Inheritance with an example.